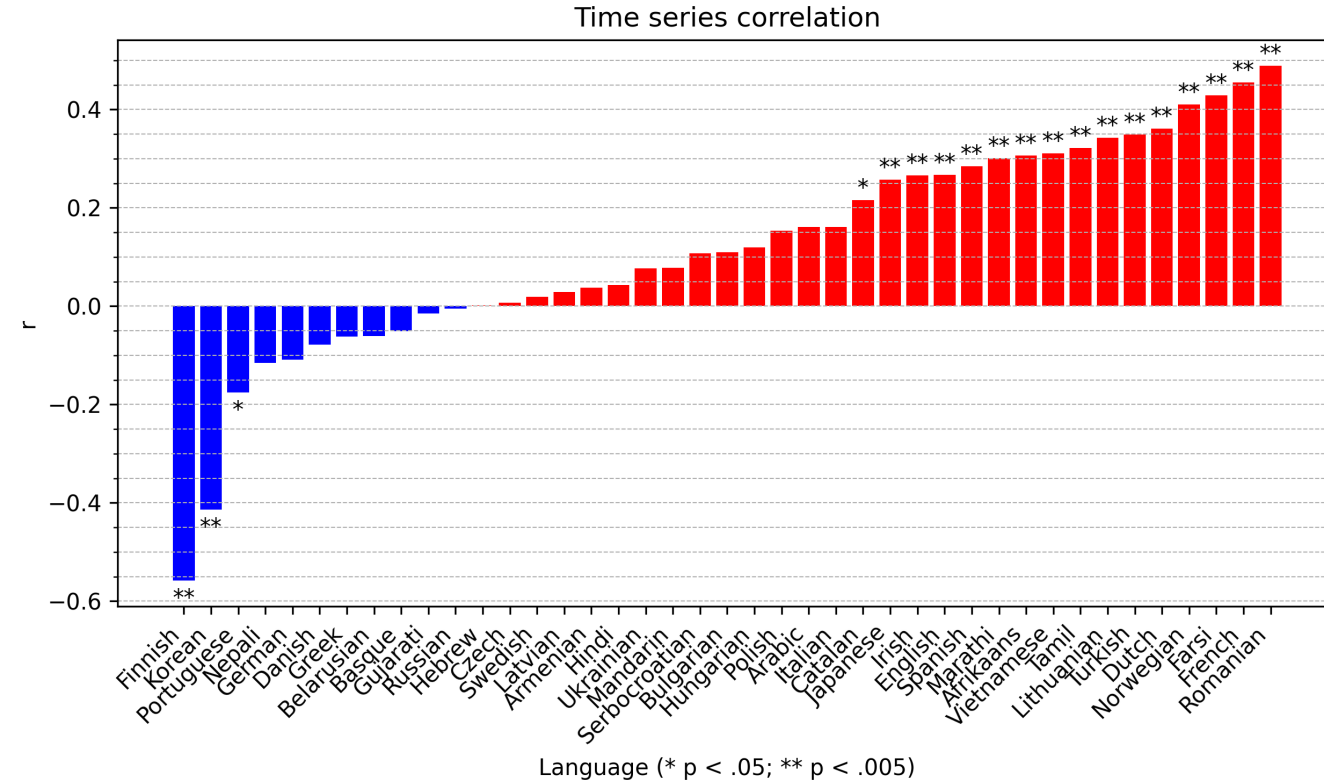


MLM – Alice project

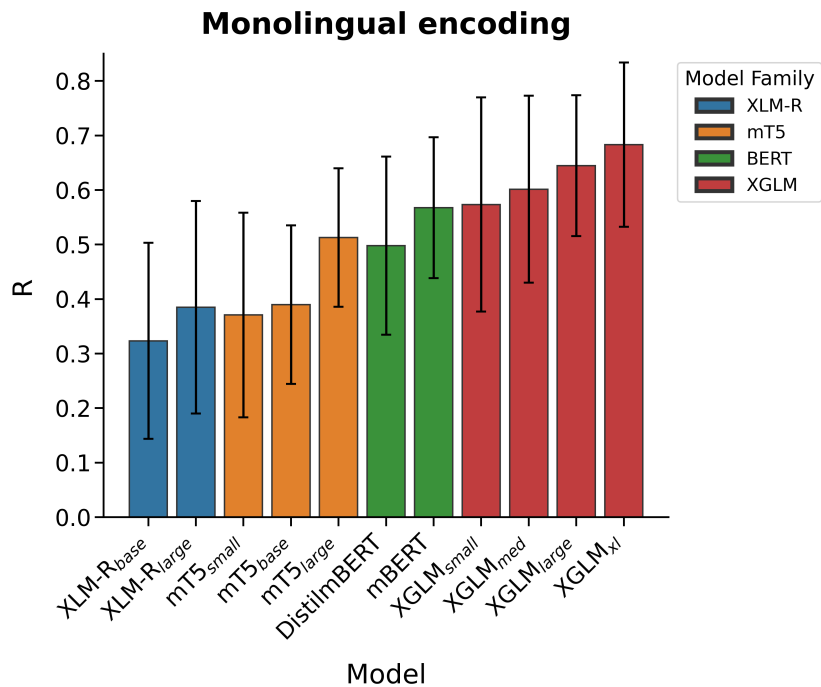
November 17 meeting

Time series correlation



- Correlations are significant in 16 languages
- Irish is not supported by whisper-timestamped
→ 15 languages

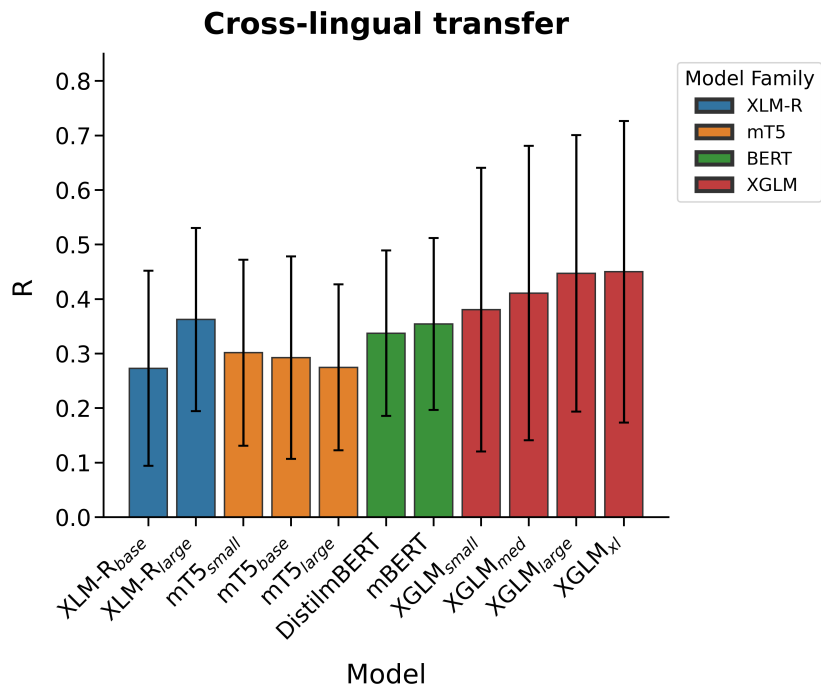
Encoding performance by model



Monolingual encoding

- Best layer for each model
- 10-fold cross-validation (random split)
- No “embedding lag”
- R values averaged across folds and languages
- Error bars: SD across languages

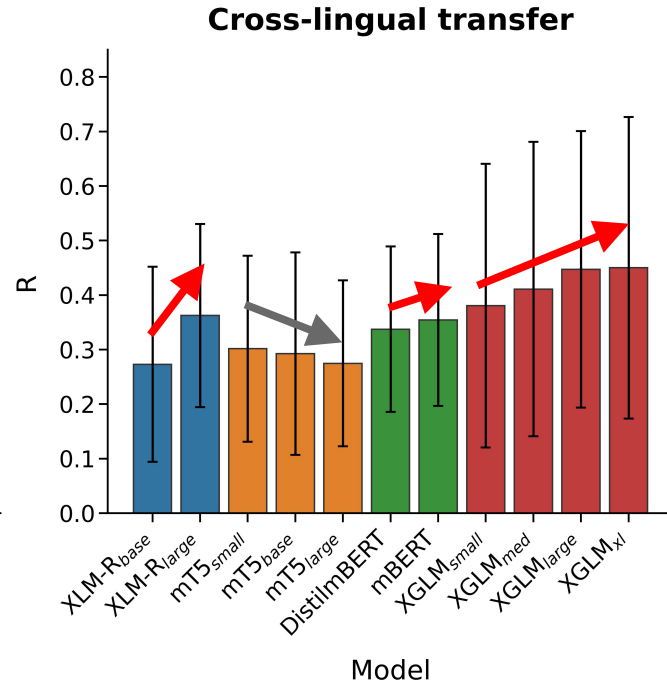
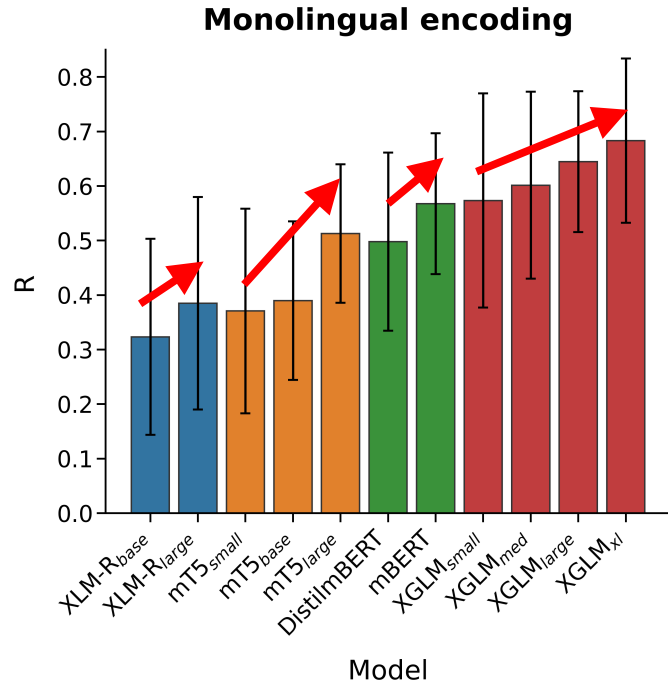
Encoding performance by model



Cross-lingual transfer

- Best layer for each model
- 10-fold cross-validation (random split)
- No “embedding lag”
- R values averaged across languages (there are no folds)
- Error bars: SD across languages

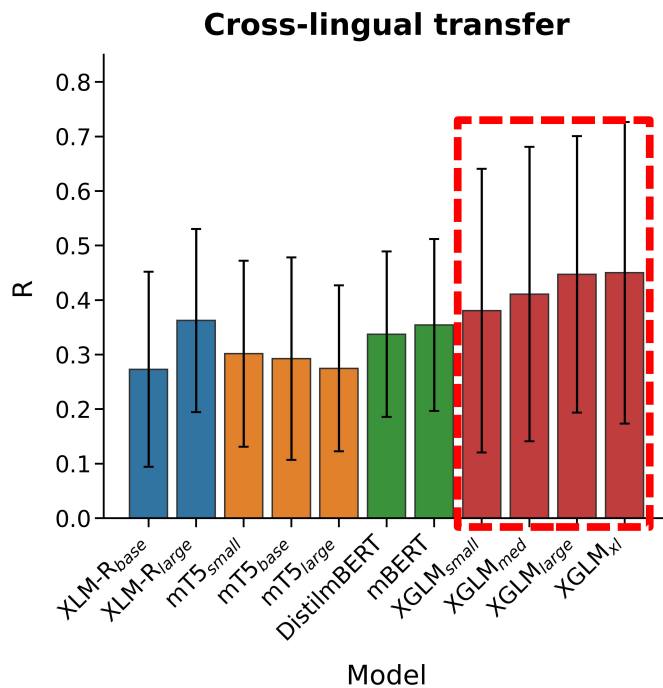
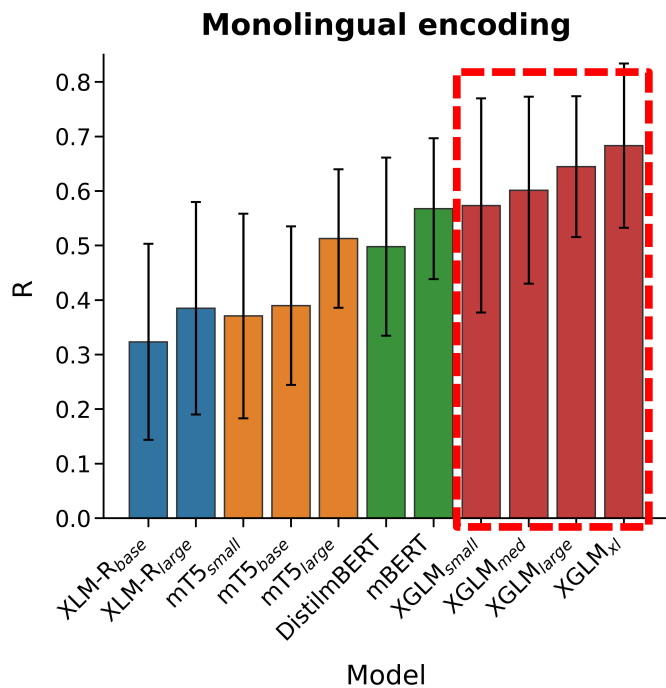
Model capacity



Larger-capacity models outperform smaller models (except for mT5 in cross-lingual transfer)

Cfr. Schrimpf et al. 2021

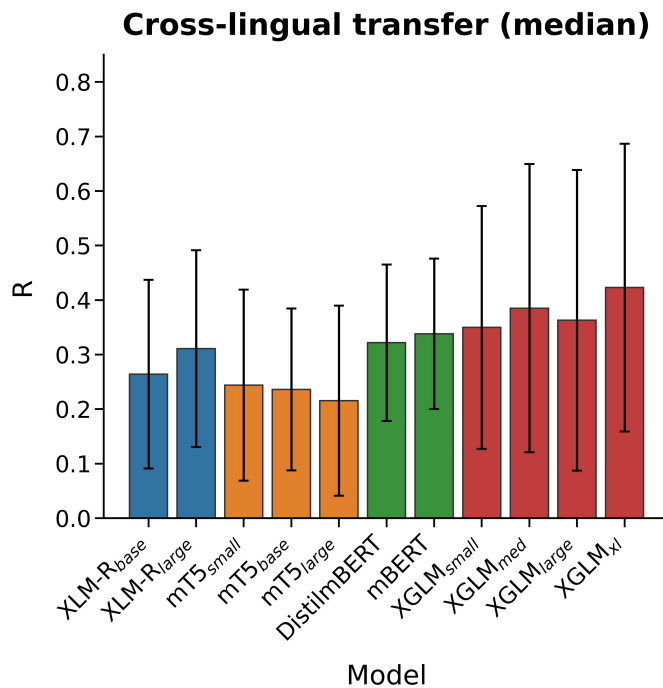
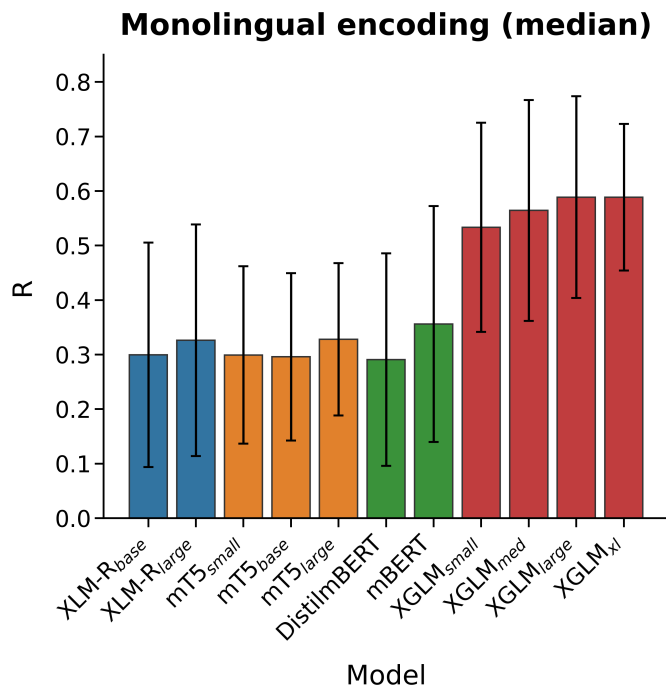
Auto-regressive vs. bidirectional



Auto-regressive
language models
outperform the
bidirectional ones

Cfr. Schrimpf et al. 2021

Median layer

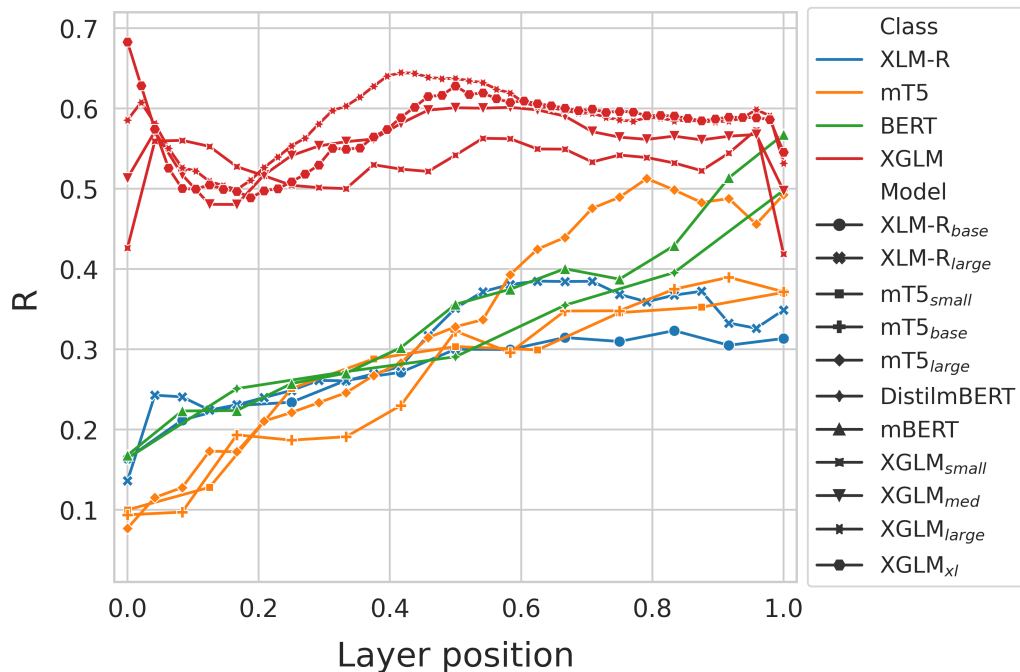


The results are similar when considering the median (as opposed to best) layer

- Bigger > smaller
- Causal > bidirectional

Encoding performance by layer

Monolingual encoding by layer

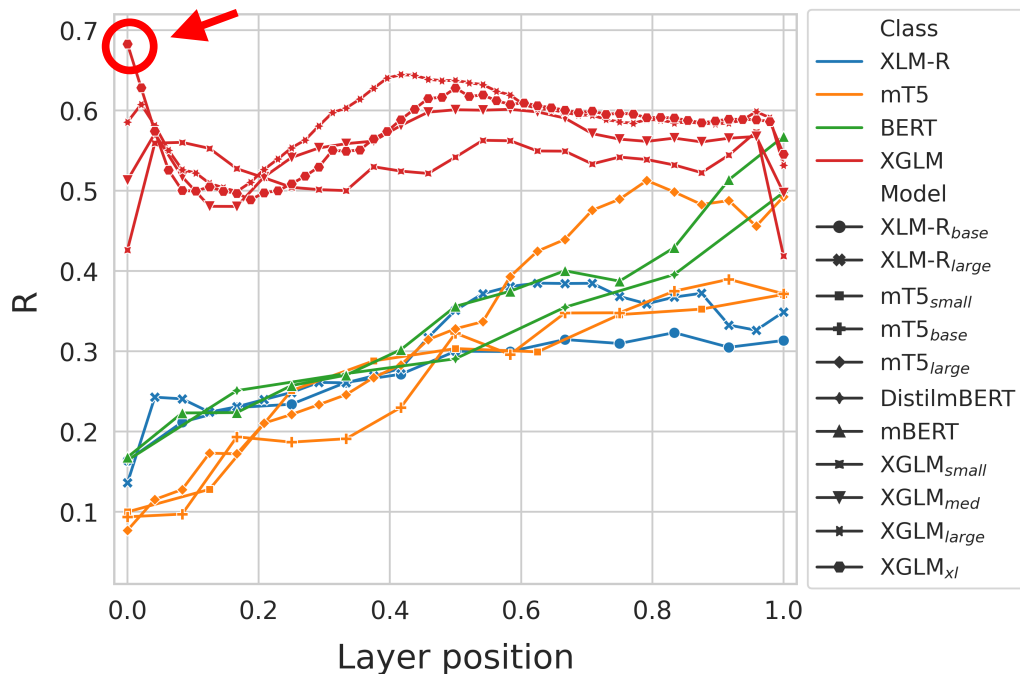


Monolingual encoding

- Middle-to-deep layers achieve the strongest predictivity
- Intermediate layers work better for the auto-regressive models
- Deep layers work better for the bidirectional models

Encoding performance by layer

Monolingual encoding by layer

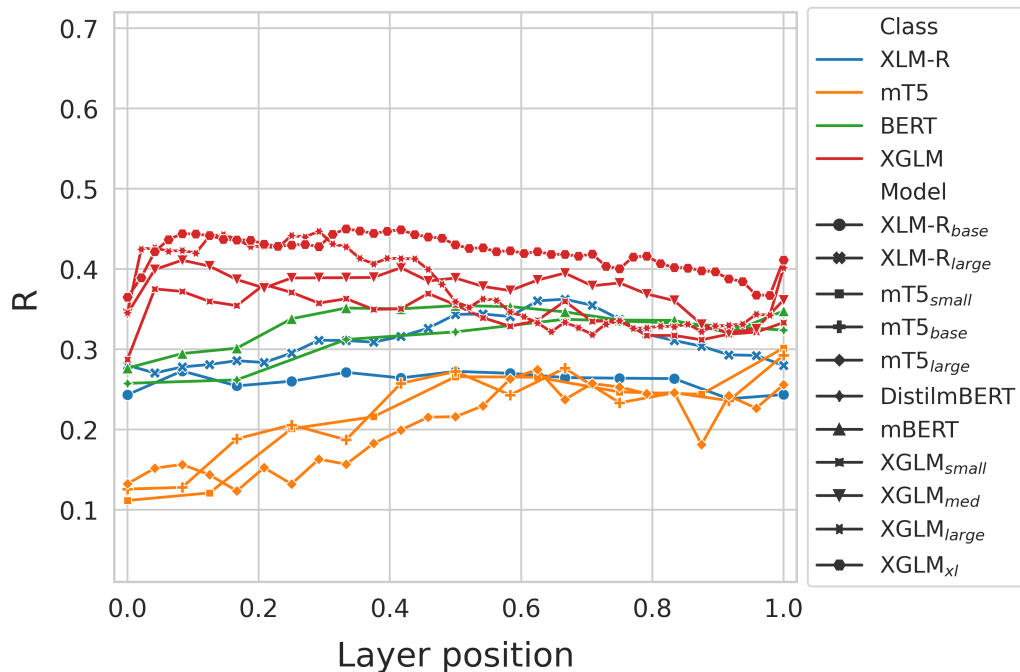


Monolingual encoding

- Middle-to-deep layers achieve the strongest predictivity
- XGLM_{xl} achieves the strongest predictivity in the embedding layer
- Deep layers work better for the bidirectional models

Encoding performance by layer

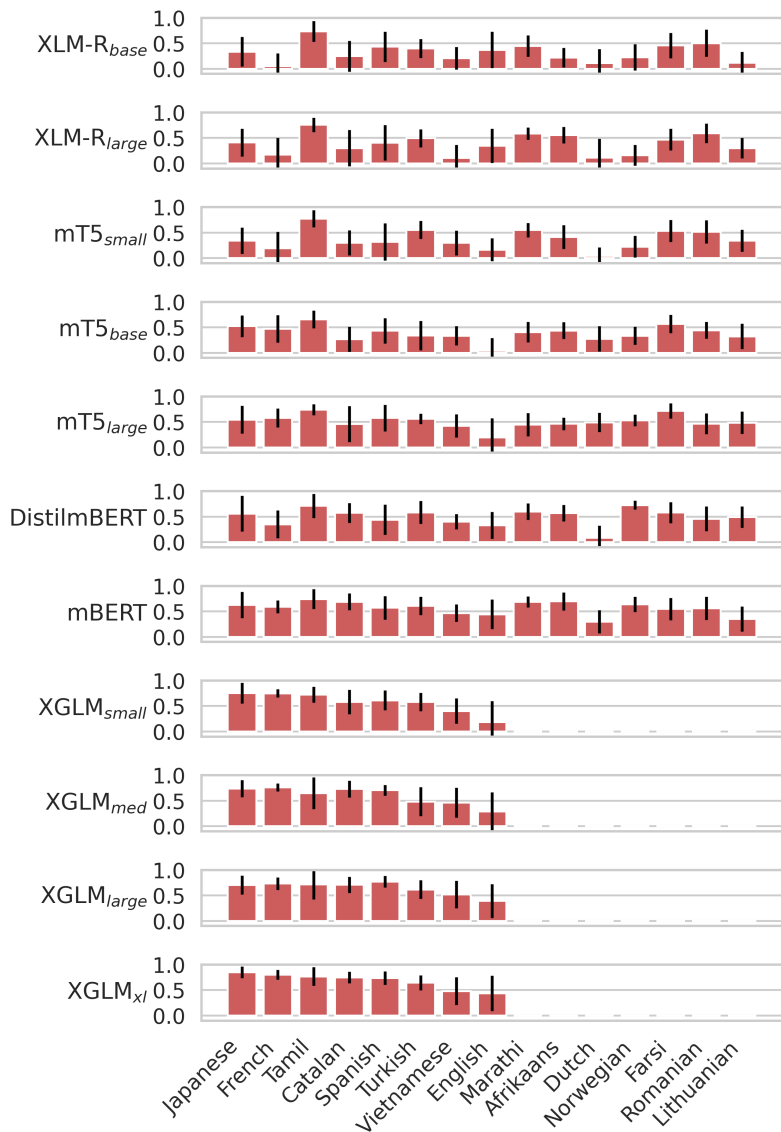
Cross-lingual transfer by layer



Multilingual encoding

- Intermediate layers achieve the strongest predictivity
 - Early-intermediate layers work better for the auto-regressive models
 - Intermediate-deep layers work better for the bidirectional models

Monolingual encoding

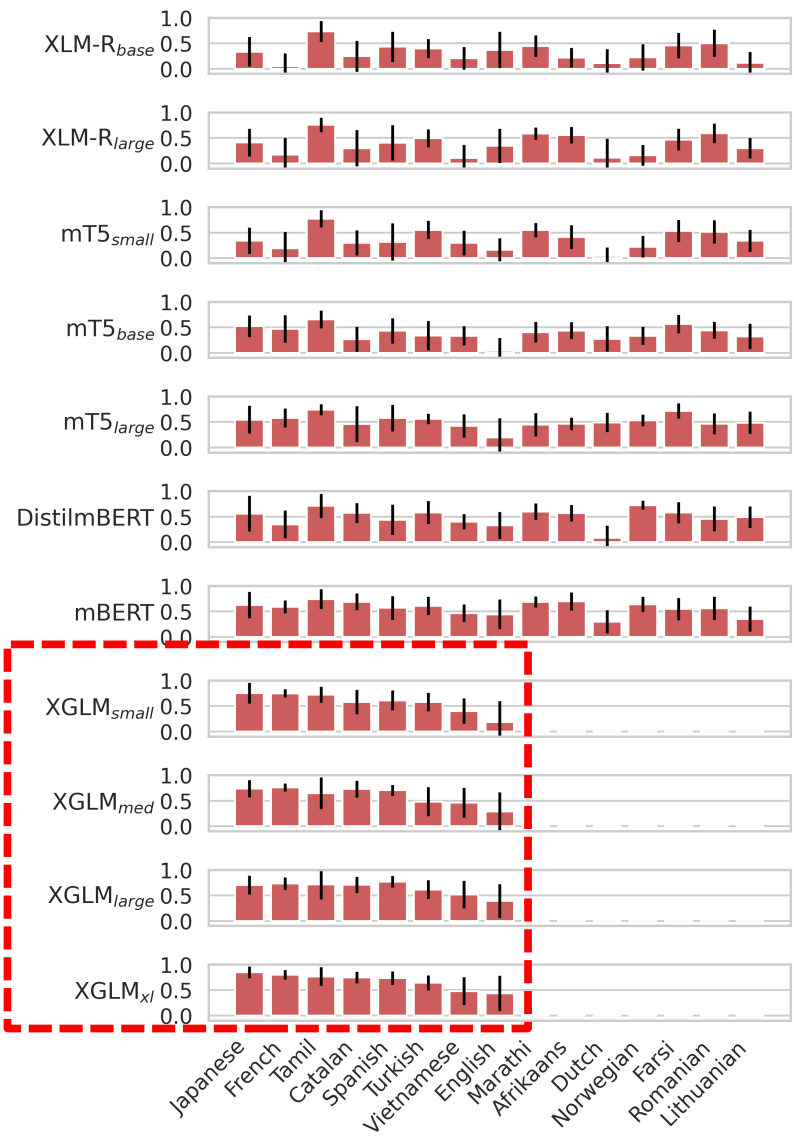


Performance by language

Monolingual encoding

- Performance obtained with the best layer (overall)
- Error bars indicate the SD across folds

Monolingual encoding

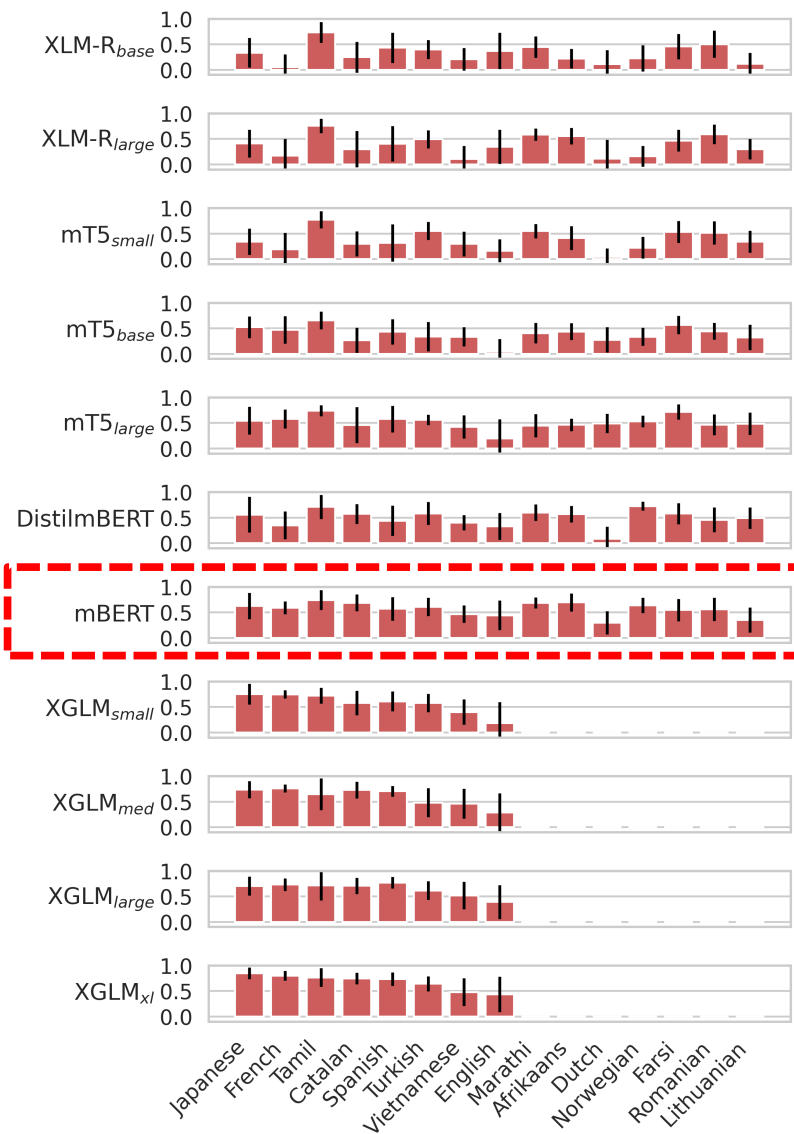


Performance by language

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- Performance obtained with the best layer (overall)
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- Auto-regressive models obtain a relatively strong performance across all 8 languages

Monolingual encoding

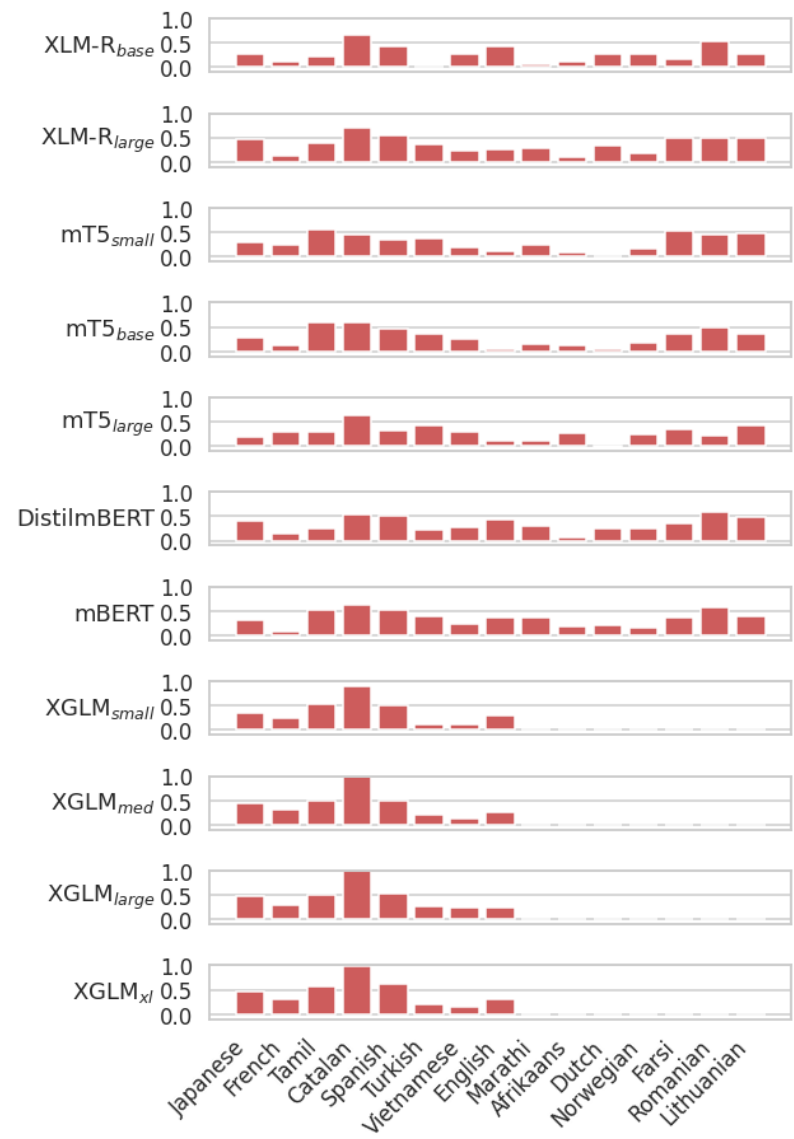


Performance by language

Monolingual encoding

- Performance obtained with the best layer (overall)
- Error bars indicate the SD across folds
- Auto-regressive models obtain a relatively strong performance across all 8 languages
- mBERT achieves relatively strong performance scores across all 15 languages

Cross-lingual transfer

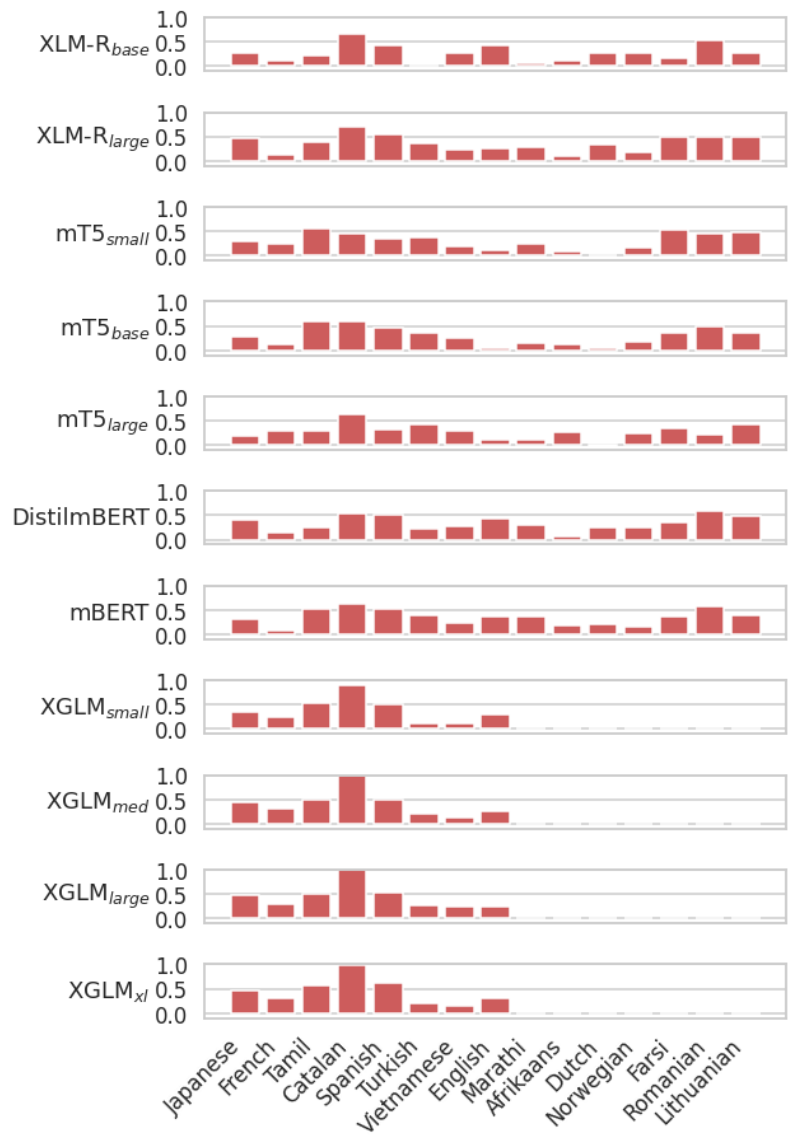


Performance by language

Multilingual encoding

For each language $l \in L$, an encoding model is fitted in all languages m in L excluding l ($m \in L \setminus \{l\}$), and tested on l .

Cross-lingual transfer



Performance by language

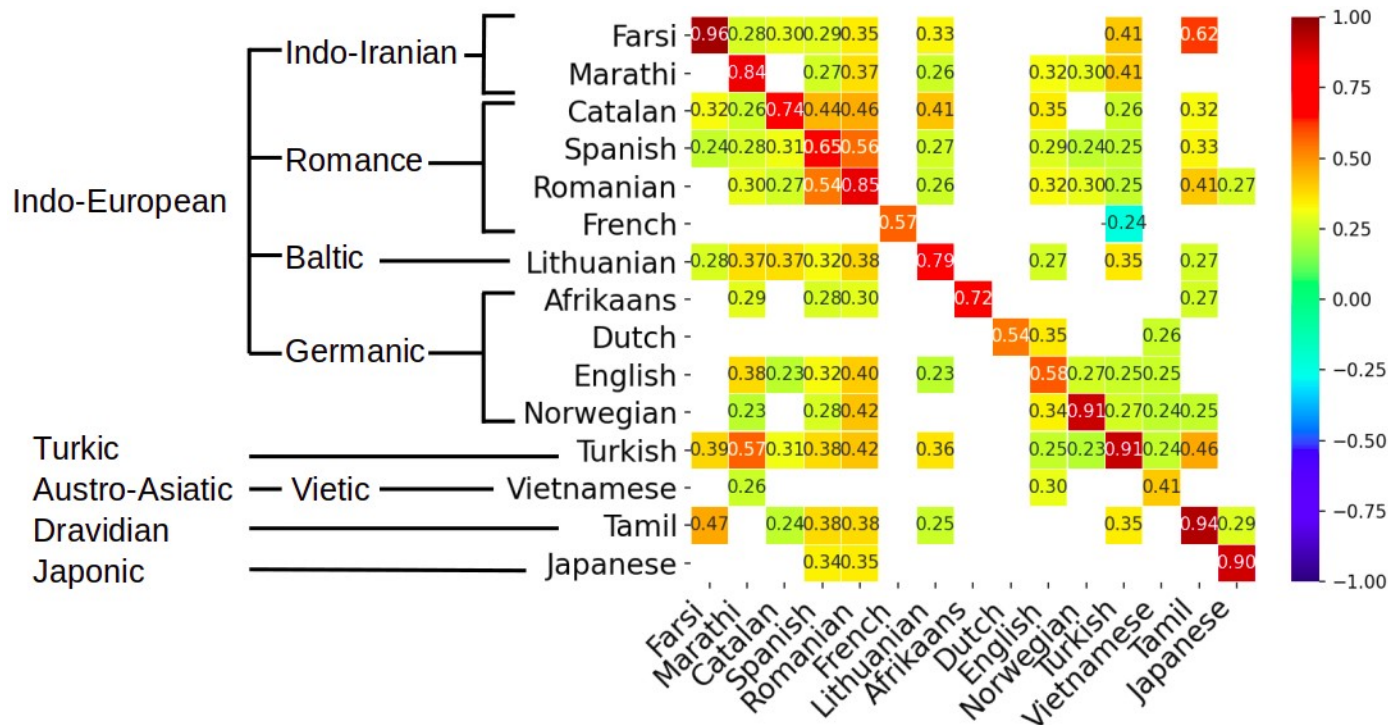
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Alternative:

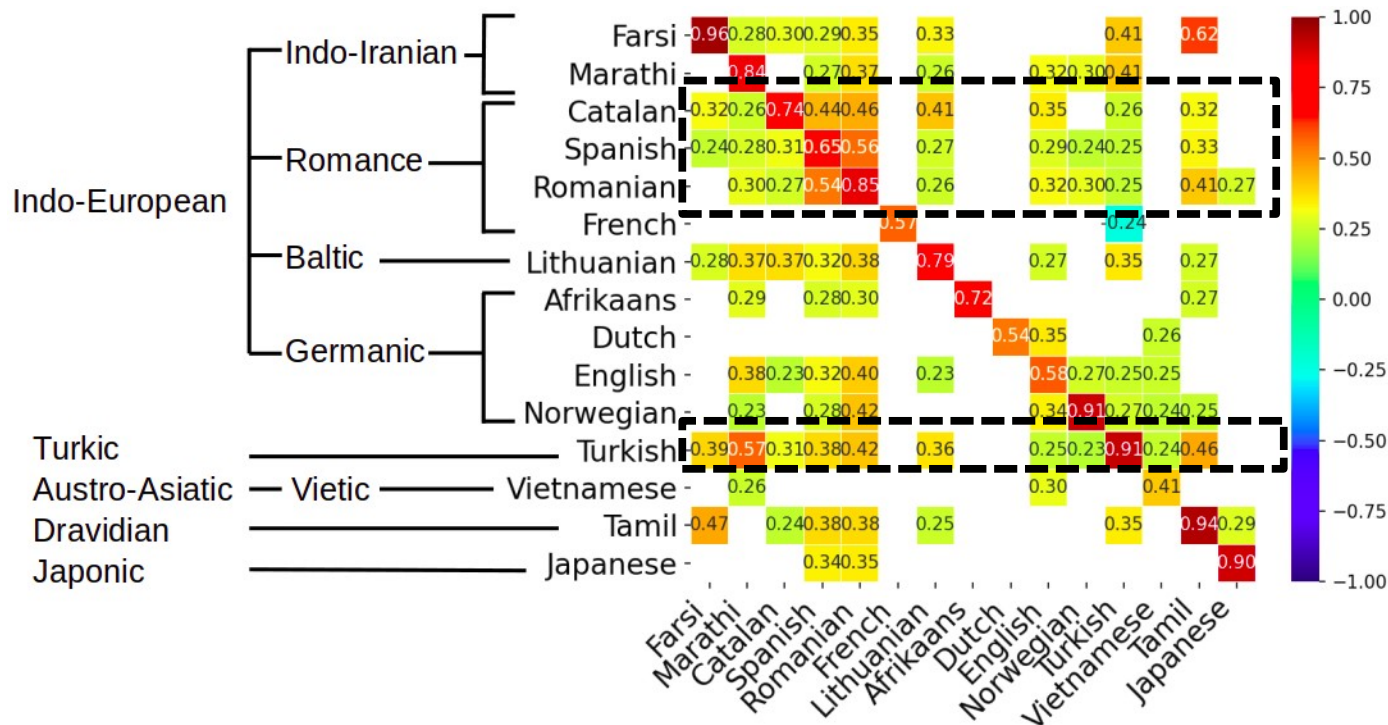
- Fit an encoding model in each language
- Transfer it zero-shot to all the other languages

Single training language



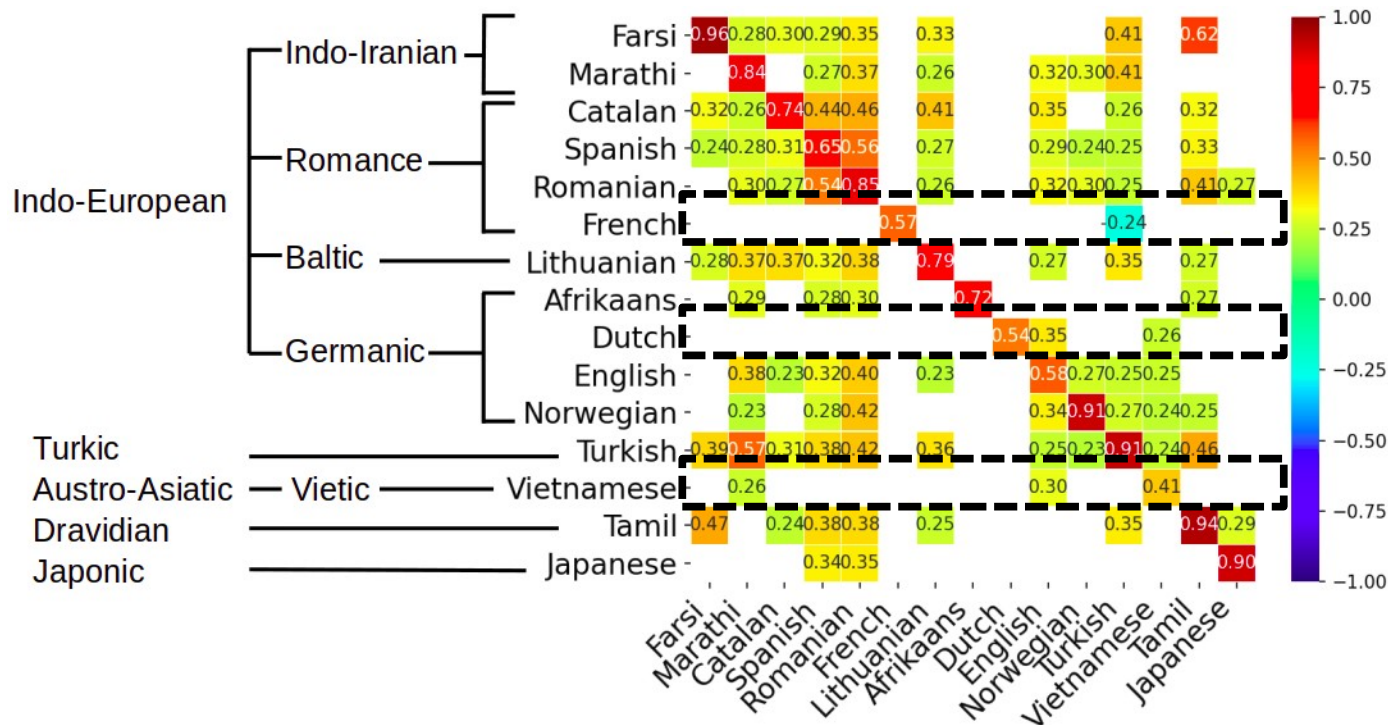
- Rows: how well a model trained in a language transfers to the other
- No clear typological pattern

Single training language



- Rows: how well a model trained in a language transfers to the other
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- Some languages generally support well the transfer
 - Turkish, Catalan, Spanish, Romanian

Single training language

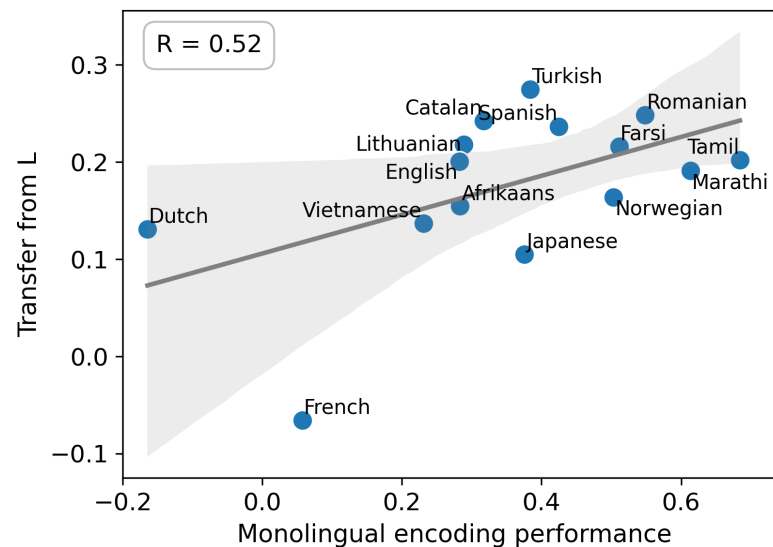


- Rows: how well a model trained in a language transfers to the other
- No clear typological pattern
- Some languages generally support well the transfer
 - Turkish, Catalan, Spanish, Romanian
- Other languages don't
 - French, Dutch, Vietnamese

Encoding and transfer performance

Transfer is generally better for some languages than for others.

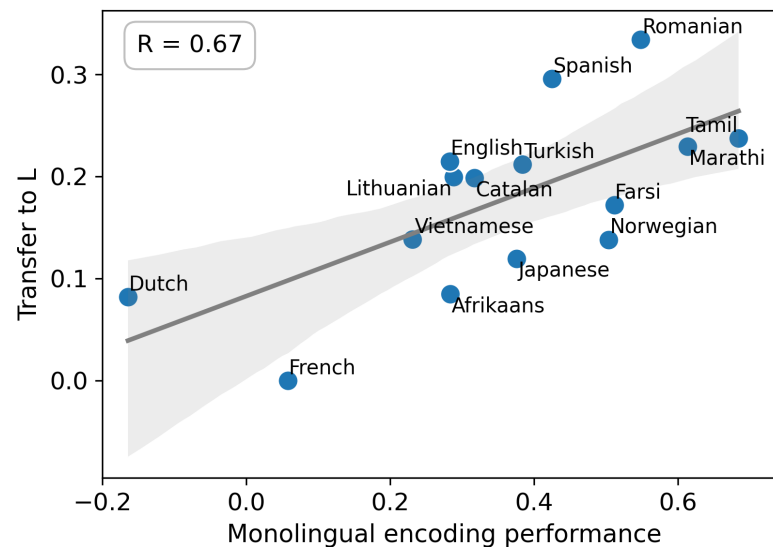
- Relationship between monolingual encoding score in a given language l and:
 - **How well the models trained on l transfers to other languages**
 - How well the models trained on other languages transfer to l



Encoding and transfer performance

Transfer is generally better for some languages than for others.

- Relationship between monolingual encoding score in a given language I and:
 - How well the models trained on I transfers to other languages
 - **How well the models trained on other languages transfer to I**



Open questions

How to evaluate statistical significance

- Paired samples permutation test?
 - Baseline where the dependent variable is randomly shuffled
 - Compare the encoding performance between target and randomized model by doing fold-wise permutation

	Model	Shuffled
Fold 1	0.46	0.10
Fold 2	0.31	0.02
Fold 3	0.28	0.04
Fold 4	0.40	0.01
Fold 5	0.63	0.15
Fold 6	0.11	0.12
Fold 7	0.55	0.03
Fold 8	0.42	0.19
Fold 9	0.64	0.00
Fold 10	0.08	0.07

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$$p = | \Delta > \Delta_0 | / N_{\text{perm}}$$

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Open questions

Extending the drive/suppress framework

- Do we still want to do it?
 - The models are already transferred to new languages, stimuli, and participants
- If we want to do it, what do we want to test?
 - Surprisal, grammaticality, plausibility, mental state, valence, etc?

Open questions

Relating the language models' performance with the prediction of fMRI activity

- (Pseudo)perplexity
 - Cfr. Schrimpf et al. 2021
- Semantic similarity
 - Existing similarity datasets (e.g., MultiSimLex) have limited coverage
 - We could use a multilingual lexicon (e.g., word2word), and see how a model can retrieve translations from and to each language

Focus on a subset of models (BERT, XGLM_{xl})?